Vulnerability Assessment

164.47.47.11

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Target Details:

The target is a set of network infrastructure located at 164.47.47.11 externally and 172.16.34.129 internally, supporting various systems for the Colorado Community College system at the ACC Strum campus in Denver, CO. The system belongs to Colorado Community College, and the contact listed in the SSL certificate is Katherine Stevenson, reachable by email at Katherine.Stevenson@cccs.edu. The main host we are targeting is an ESXi VMware server running version 6.7, which manages virtual machines for educational purposes.

Set of Internal IPS

- 172.16.34.1
- 172.16.34.2
- 172.16.34.3
- 172.16.34.4
- 172.16.34.5
- 172.16.34.7
- 172.16.34.10
- 172.16.34.24
- 172.16.34.30
- 172.16.34.130
- 172.16.34.131
- 172.16.34.209

Rules of Engagement:

All testing was done with the authorization of the ACC Staff, We received permission from Tyler Dice and James Smith for the following.

- Allowed basic network discovery and basic web discovery.
- Authorized tools authorized were Nessus, Greenbone, Zap, and Nmap.
- Brute-forcing data and data manipulation were disallowed.
- Testing was only allowed during authorized times.

Points of Contact

Tyler Dice (CTE Lab Coordinator at ACC)

CTE Lab Coordinator at ACC

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James Smith (CTE Lab Coordinator at ACC)

CTE Lab Coordinator

James.Smith@arapahoe.edu

Erin Owens (Professor, ACC)

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Scoring:

To evaluate and categorize our findings, we will be using the CVSS v3.0 Scoring System. This will allow us to create a standardized method to assess vulnerabilities by assigning a severity score based on their potential impact and level of risk. We are presenting the risk through this scoring system, making it more streamlined visually.

Rating	Definition	Score
Info	**Informational Data, Poses No Threat. No Action Required**	**0**
Low	**Minimal threat, no immediate threat.**	**0.1 - 3.9**
Medium	**Moderate threat, could cause a breach if exploited, Requires Attention**	**4.0 - 6.9**
High	**Significant Threat, Likely to be exploited and cause damage. Action Required ASAP**	**7.0 - 8.9**
Critical	**Immediate Threat, Likely to cause significant damage and breaches. Action is required immediately and should be prioritized**	**9.0 - 10.0**

Executive Summary:

This vulnerability assessment was conducted on a VMware ESXi server located at the Arapahoe Community College STURM campus, which is part of the Colorado Community College System. The target system, with an external IP address of 164.47.47.11 and an internal IP address of 172.16.34.129, is an ESXi VMware server primarily used for educational purposes to initialize and manage virtual machines throughout various classes. This assessment was performed with authorization from Professor Erin Owens, CTE Lab Coordinators Tyler Dice and James Smith, and other staff members at ACC.

The testing, which was performed using tools such as Nessus, Greenbone, OWASP techniques, etc. identified several critical vulnerabilities and configuration issues in the target systems. The findings include outdated versions of VMware ESXi with known CVEs, outdated SSL certificates, and weak security configurations across various components of the network infrastructure.

This vulnerability assessment has identified several critical and high-severity vulnerabilities across various systems, including VMware ESXi, vCenter Server, Microsoft SQL Server, and Unix-based operating systems. These vulnerabilities expose systems to risks such as arbitrary code execution, denial of service (DoS), sensitive data exposure, and cryptographic weaknesses that could lead to data compromise. The most severe findings are related to unsupported and outdated versions of VMware ESXi, vCenter Server, and Microsoft SQL Server, all of which are subject to critical

vulnerabilities that can be exploited remotely. Additionally, several SSL/TLS vulnerabilities, including the use of weak encryption algorithms and outdated SSL protocols, further compromise system integrity and security.

We recommend the immediate upgrade of unsupported versions of VMware ESXi, vCenter Server, and Microsoft SQL Server. Critical security patches, such as those for CVE-2018-6982 and CVE-2018-6981, should be applied to VMware ESXi. SSL/TLS settings should be updated to use more secure options and disable the outdated versions. The self-signed SSL certificate should be replaced with one issued by a trusted Certificate Authority to ensure proper security and trustworthiness.

Testing Summary:

The system we tested is a server with VMware ESXi running on it. ESXi runs without an operating system, including its kernel. ESXi provides the users with a web console to be able to login and create Virtual Machines. As for our testing, we were authorized by Professor Erin Owens, he is the instructor of the class in which we operated this test. Other authorizing figures of ACC's Cyber Lab are: Tyler Dice and James Smith of Arapahoe Community College at the Sturm Collaboration Campus.

Tools used

Shodan - Used to perform a deep search of the target's publicly exposed services and systems, providing information about device types, open ports, and any potentially vulnerable software.

IPVoid - Used to check the ownership details of the SSL certificate, helping to identify the certificate issuer and the contact information associated with the domain.

Nmap - Conducted a thorough network scan to map out the open ports and services running on the target system, helping to identify areas vulnerable to exploitation.

Greenbone - Ran targeted scans on the internal and external infrastructure to detect vulnerabilities, focusing on outdated software versions and misconfigurations across the network.

Nessus - Performed in-depth vulnerability assessments to identify critical and high-risk vulnerabilities, including missing patches, weak configurations, and known CVEs affecting the target system.

ZAP - Used to scan the web applications hosted on the target server, detecting potential security issues like broken authentication mechanisms, insecure session management, and input validation flaws.

Scope:

The scope of this assessment focused on identifying vulnerabilities within the external target ip at 164.47.47 and internal ip at 172.16.34.129, which belongs to the Colorado Community College System (CCCS) located in Denver, CO.

Timeline:

External Testing was performed between September 3, and October 8, beginning with passive scanning using tools such as Shodan and IPVoid to gather preliminary information on the target and verify proper authorizations. Once the initial reconnaissance phase was complete, we proceeded to active scanning using tools like Nessus and ZAP to further identify potential vulnerabilities. After the external tests, we fingerprinted and footprinted our target using Nmap so we could start internal testing which took place between November 19, and December 5, by using tools such as Greenbone, and Nessus scan.

Methodology:

The primary objective of our internal testing methodology was to gather information and actively search for vulnerabilities. Some of our methods include using Shodan, and IPVoid to collect publicly available information about the IP address such as: the hostname, open ports geographical location and associated organizations. Using this info, we gathered info to identify potential entry points for attacks. Using the other tools at our disposal such as Greenbone and Nessus we scanned for more vulnerabilities and prioritized them based on their severity and the potential impact to Arapahoe Community College and its staff. Using all this knowledge, we documented and outlined these severities and made recommendations and mitigation strategies all in accordance to the rules that were set by our points of contact.

Findings and Recommendations:

External

164.47.47.11- ESXi VMware Server (External IP)

Critical	High	Medium	Low	Info
1	0	2	0	20

Vulnerability	Description	Score
CVE-2024-22252, CVE-2024-22253, CVE-2024-22254, CVE-2024-22255	The version of VMware ESXi installed on the remote host is prior to 7.0 Update 3p, 8.0 prior to 8.0 Update 1d, or 8.0 prior to 8.0 Update 2b	**9.3**

Recommendations: Upgrade to VMware ESXi 7.0 Update 3p, 8.0 Update 1d, or 8.0

Update 2b or later

Vulnerability	Description	Score
CVE-2024-37086	The version of VMware ESXi installed on the remote host is prior to 7.0 Update 3q or 8.0 prior to 8.0 Update 3. It is, therefore, affected by an out-of-bounds vulnerability	**6.8**

Recommendations: Upgrade to VMware ESXi 7.0 Update 3p, 8.0 Update 1d, or 8.0

Update 2b or later

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service)

Internal

172.16.34.129- ESXi VMware Server (Internal IP)

Critical	High	Medium	Low	Info
1	0	2	0	29

Vulnerability	Description	Score
CVE-2024-22255, CVE-2024-22254, CVE-2024-22253, CVE-2024-22252	The version of VMware ESXi installed on the remote host is prior to 7.0 Update 3p, 8.0 prior to 8.0 Update 1d, or 8.0 prior to 8.0 Update 2b	**9.3**

Recommendations: Upgrade to VMware ESXi 7.0 Update 3p, 8.0 Update 1d, or 8.0

Update 2b or later

Vulnerability	Description	Score
CVE-2024-37086	The version of VMware ESXi installed on the remote host is prior to 7.0 Update 3q or 8.0 prior to 8.0 Update 3. It is, therefore, affected by an out-of-bounds vulnerability	**6.8**

Recommendations: Upgrade to VMware ESXi 7.0 Update 3p, 8.0 Update 1d, or 8.0

Update 2b or later

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

172.16.34.1- Palo Alto Networks

Critical	High	Medium	Low	Info
0	1	6	4	39

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium-strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2020-11023, CVE-2020-11022	According to the self-reported version in the script, the version of JQuery hosted on the remote web server is greater than or equal to 1.2 and prior to 3.5.0. It is, therefore, affected by multiple cross site scripting vulnerabilities	**6.1**

Recommendations: Upgrade to JQuery version 3.5.0 or later

Vulnerability	Description	Score
CVE-2023-48795	The remote SSH server is vulnerable to a man-in-the-middle prefix truncation weakness known as Terrapin. This can allow a remote, man-in-the-middle attacker to bypass integrity checks and downgrade the connection's security	**5.9**

Recommendations: Contact the vendor for an update with the strict key exchange countermeasures or disable the affected algorithms

Vulnerability	Description	Score
SSH Weak Algorithms Supported	The remote SSH Server is configured to use the Arcfour stream cipher or no cipher at all. RFC 4253 advises against using Arcfour due to an issue with weak keys	**4.3**

Recommendations: Contact the vendor or consult product documentation to remove the weak ciphers

172.16.34.2- Netlab

Critical	High	Medium	Low	Info
0	0	1	0	29

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

172.16.34.3- Cisco Switch

Critical	High	Medium	Low	Info
0	1	1	4	17

Vulnerability	Description	Score
CVE-2001-1473, CVE-2001-0572, CVE-2001-0361	The remote SSH daemon supports connections made using the version 1.33 and/or 1.5 of the SSH protocol	**7.3**

Recommendations: Disable compatibility with version 1 of the SSH

Vulnerability	Description	Score
Unencrypted Telnet Server	The remote host is running a Telnet server over an unencrypted channel	**6.5**

Recommendations: Disable the Telnet service and use SSH instead

172.16.34.4- ESXi VMware Server

Critical	High	Medium	Low	Info
2	2	8	1	36

Vulnerability	Description	Score
Unix Operating System Unsupported Version Detection	The operating system running on the remote host is no longer supported	**10.0**

Recommendations: Upgrade to a version of the Unix operating system that is currently supported

Vulnerability	Description	Score
VMware ESX /ESXi Unsupported Version Detection	The remote host is running an unsupported version of a virtualization application	**10.0**

Recommendations: Upgrade to a version of VMware ESX / ESXi that is currently supported

Vulnerability	Description	Score
CVE-2018-6982, CVE-2018-6981	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to multiple vulnerabilities. Leveraging the most severe of these vulnerabilities could allow an attacker to execute arbitrary code on the host from the security context of an unprivileged user on the guest system	**8.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
CVE-2018-6974	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to an out-of-bounds read vulnerability in SVGA devices. An attacker with access to a guest system may be able to execute code on the host system by leveraging this vulnerability	**8.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

172.16.34.5- ESXi VMware Server

Critical	High	Medium	Low	Info
2	1	5	1	37

Vulnerability	Description	Score
Unix Operating System Unsupported Version Detection	The operating system running on the remote host is no longer supported	**10.0**

Recommendations: Upgrade to a version of the Unix operating system that is currently supported

Vulnerability	Description	Score
VMware ESX /ESXi Unsupported Version Detection	The remote host is running an unsupported version of a virtualization application	**10.0**

Recommendations: Upgrade to a version of VMware ESX / ESXi that is currently supported

Vulnerability	Description	Score
CVE-2018-6982, CVE-2018-6981	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to multiple vulnerabilities. Leveraging the most severe of these vulnerabilities could allow an attacker to execute arbitrary code on the host from the security context of an unprivileged user on the guest system	**8.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
CVE-2019-5519, CVE-2019-5518	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to multiple vulnerabilities	**6.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

172.16.34.7- ESXi VMware Server

Critical	High	Medium	Low	Info
2	2	8	1	32

Vulnerability	Description	Score
Unix Operating System Unsupported Version Detection	The operating system running on the remote host is no longer supported	**10.0**

Recommendations: Upgrade to a version of the Unix operating system that is currently supported

Vulnerability	Description	Score
VMware ESX /ESXi Unsupported Version Detection	The remote host is running an unsupported version of a virtualization application	**10.0**

Recommendations: Upgrade to a version of VMware ESX / ESXi that is currently supported

Vulnerability	Description	Score
CVE-2018-6982, CVE-2018-6981	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to multiple vulnerabilities. Leveraging the most severe of these vulnerabilities could allow an attacker to execute arbitrary code on the host from the security context of an unprivileged user on the guest system	**8.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
CVE-2019-5519, CVE-2019-5518	The remote VMware ESXi host is version 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to multiple vulnerabilities	**6.8**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
CVE-2018-6972	The remote VMware ESXi host is version 5.5, 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to a denial of service vulnerability	**6.5**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2018-3646	The remote VMware ESXi host is version 5.5, 6.0, 6.5, or 6.7 and is missing a security patch. It is, therefore, vulnerable to a speculative execution side channel attack known as L1 Terminal Fault	**5.6**

Recommendations: Apply the appropriate patch as referenced in the vendor advisory

172.16.34.10- VMware vCenter Server

Critical	High	Medium	Low	Info
2	2	5	0	37

Vulnerability	Description	Score
CVE-2017-5641	The version of VMware vCenter Server installed on the remote host is 6.0.x prior to 6.0u3b or 6.5.x prior to 6.5c. It is, therefore, affected by a flaw in FlexBlazeDS when processing AMF3 messages due to allowing the instantiation of arbitrary classes when deserializing objects	**9.8**

Recommendations: Upgrade to VMware vCenter Server version 6.0u3b (6.0.0 build-5326177) / 6.0u3b on Windows (6.0.0 build-5318198) / 6.5.0c (6.5.0 build-5318112) or later. Alternatively, apply the vendor-supplied workaround

Vulnerability	Description	Score
CVE-2016-7460, CVE-2016-7459	The version of VMware vCenter Server installed on the remote host is 5.5.x prior to 5.5u3e or 6.0.x prior to 6.0u2a. It is, therefore, affected by multiple XML external entity (XXE) vulnerabilities	**9.1**

Recommendations: Upgrade to VMware vCenter Server version 5.5.u3e (5.5.0 build-4180646) / 6.0u2a (6.0.0 build-4541947) or later

Vulnerability	Description	Score
CVE-2019-5534, CVE-2019-5532, CVE-2019-5531	The version of VMware vCenter Server installed on the remote host is 6.0 prior to U3j, 6.5 prior to U3, or 6.7 prior to U3, and is, therefore, affected by the following vulnerabilities	**7.7**

Recommendations: Upgrade to VMware vCenter Server 6.0 U3j, 6.5 U3, or 6.7 U3 or later

Vulnerability	Description	Score
CVE-2017-4928, CVE-2017-4927	The version of VMware vCenter Server installed on the remote host is 5.5.x prior to 5.5u3f, 6.0.x prior to 6.0u3c, or 6.5.x prior to 6.5u1. It is, therefore, affected by multiple vulnerabilities	**7.5**

Recommendations: Upgrade to VMware vCenter Server version 5.5.u3f (5.5.0 build-6516310) / 6.0u3c (6.0.0 build-7037393) / 6.5u1 (6.5.0 build-5973321) or later

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2017-5715	The version of VMware vCenter Server installed on the remote host is 5.5.x prior to 5.5U3g, 6.0.x prior to 6.0U3d, or 6.5.x prior to 6.5U1e	**5.6**

Recommendations: Upgrade to VMware vCenter Server version 5.5.U3g (5.5.0 build-7460778) / 6.0U3d (6.0.0 build-7464194) / 6.5U1e (6.5.0 build-7515524) or later

Vulnerability	Description	Score
CVE-2018-3646	The version of VMware vCenter Server installed on the remote host is 5.5.x prior to 5.5u3j, 6.0.x prior to 6.0u3h, 6.5.x prior to 6.5u2c, or 6.7.x prior to 6.7.0d	**5.6**

Recommendations: Upgrade to the version, or later, referenced in the vendor advisory

172.16.34.24- Windows Server 2016

Critical	High	Medium	Low	Info
2	3	12	3	76

Vulnerability	Description	Score
Microsoft SQL Server Unsupported Version Detection	According to its self-reported version number, the installation of Microsoft SQL Server on the remote host is no longer supported	**10.0**

Recommendations: Upgrade to a version of Microsoft SQL Server that is currently supported

Vulnerability	Description	Score
SSL Version 2 and 3 Protocol Detection	The remote service accepts connections encrypted using SSL 2.0 and/or SSL 3.0. These versions of SSL are affected by several cryptographic flaws	**9.8**

Recommendations: Consult the application's documentation to disable SSL 2.0 and 3.0. Use TLS 1.2 (with approved cipher suites) or higher instead

Vulnerability	Description	Score
CVE-2005-4900, CVE-2004-2761	The remote service uses an SSL certificate chain that has been signed using a cryptographically weak hashing algorithm (e.g. MD2, MD4, MD5, or SHA1). These signature algorithms are known to be vulnerable to collision attacks	**7.5**

Recommendations: Contact the Certificate Authority to have the SSL certificate reissued

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2015-2808, CVE-2013-2566	The remote host supports the use of RC4 in one or more cipher suites. The RC4 cipher is flawed in its generation of a pseudo-random stream of bytes so that a wide variety of small biases are introduced into the stream, decreasing its randomness	**5.9**

Recommendations: Reconfigure the affected application, if possible, to avoid use of RC4 ciphers. Consider using TLS 1.2 with AES-GCM suites subject to browser and web server support

Vulnerability	Description	Score
SMB Signing not required	Signing is not required on the remote SMB server. An unauthenticated, remote attacker can exploit this to conduct man-in-the-middle attacks against the SMB server	**5.3**

Recommendations: Enforce message signing in the host's configuration. On Windows, this is found in the policy setting 'Microsoft network server: Digitally sign communications (always)'. On Samba, the setting is called 'server signing'

Vulnerability	Description	Score
Terminal Services Doesn't Use Network Level Authentication (NLA) Only	The remote Terminal Services is not configured to use Network Level Authentication (NLA) only. NLA uses the Credential Security Support Provider (CredSSP) protocol to perform strong server authentication either through TLS/SSL or Kerberos mechanisms, which protect against man-in-the-middle attacks	**4.0**

Recommendations: Enable Network Level Authentication (NLA) on the remote RDP server. This generally done on the 'Remote' table of the 'System' settings on Windows

172.16.34.30- HPE ProLiant ML110 PS NA Server

Critical	High	Medium	Low	Info
2	4	3	4	35

Vulnerability	Description	Score
CVE-2020-11914, CVE-2020-11912, CVE-2020-11911, CVE-2020-11907, CVE-2020-11906, CVE-2020-11900, CVE-2020-11898, CVE-2020-11896	Multiple security vulnerabilities have been identified in Integrated Lights-Out firmware generation 3 (iLO 3) prior to version 1.93, generation 4 (iLO 4) prior to version 2.75, and generation 5 (iLO 5) prior to version 2.18. Superdome generation 4 versions prior to 1.64 and Moonshot/Edgeline generation 5 versions prior to 2.30 are also vulnerable. The vulnerabilities could be remotely exploited to execute code, cause denial of service, and expose sensitive information	**10.0**

Recommendations: Upgrade to HP iLO 3 firmware version 1.93 or later, iLO 4 firmware version 2.75 or later, or iLO 4 for Superdome version 1.64 or later, or HP iLO 5 firmware version 2.18 or HP Moonshot/Edgeline iLO 5 to version 2.30 or later.

Vulnerability	Description	Score
CVE-2022-23704	A potential security vulnerability has been identified in Integrated Lights-Out 4 (iLO 4). The vulnerability could allow remote Denial of Service	**7.5**

Recommendations: Upgrade firmware to 2.80 or later

Vulnerability	Description	Score
CVE-2013-4786	The remote host supports IPMI v2.0. The Intelligent Platform Management Interface (IPMI) protocol is affected by an information disclosure vulnerability due to the support of RMCP+ Authenticated Key-Exchange Protocol (RAKP) authentication	**7.5**

Recommendations: There is no patch for this vulnerability; it is an inherent problem with the specification for IPMI v2.0. Suggested mitigations include :

- Disabling IPMI over LAN if it is not needed

- Using strong passwords to limit the successfulness of off-line dictionary attacks

- Using Access Control Lists (ACLs) or isolated networks to limit access to your IPMI management interfaces

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium-strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS

1.1

172.16.34.130- Palo Alto Networks

Critical	High	Medium	Low	Info
0	1	6	4	39

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium-strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2020-11023, CVE-2020-11022	According to the self-reported version in the script, the version of JQuery hosted on the remote web server is greater than or equal to 1.2 and prior to 3.5.0. It is, therefore, affected by multiple cross site scripting vulnerabilities	**6.1**

Recommendations: Upgrade to JQuery version 3.5.0 or later

Vulnerability	Description	Score
CVE-2023-48795	The remote SSH server is vulnerable to a man-in-the-middle prefix truncation weakness known as Terrapin. This can allow a remote, man-in-the-middle attacker to bypass integrity checks and downgrade the connection's security	**5.9**

Recommendations: Contact the vendor for an update with the strict key exchange countermeasures or disable the affected algorithms

Vulnerability	Description	Score
SSH Weak Algorithms Supported	The remote SSH Server is configured to use the Arcfour stream cipher or no cipher at all. RFC 4253 advises against using Arcfour due to an issue with weak keys	**4.3**

Recommendations: Contact the vendor or consult product documentation to remove the weak ciphers

172.16.34.131- Cisco Router

Critical	High	Medium	Low	Info
1	3	6	5	34

Vulnerability	Description	Score
SSL Version 2 and 3 Protocol Detection	The remote service accepts connections encrypted using SSL 2.0 and/or SSL 3.0. These versions of SSL are affected by several cryptographic flaws	**9.8**

Recommendations: Consult the application's documentation to disable SSL 2.0 and 3.0.

Use TLS 1.2 (with approved cipher suites) or higher instead

Vulnerability	Description	Score
CVE-2005-4900, CVE-2004-2761	The remote service uses an SSL certificate chain that has been signed using a cryptographically weak hashing algorithm (e.g. MD2, MD4, MD5, or SHA1). These signature algorithms are known to be vulnerable to collision attacks. An attacker can exploit this to generate another certificate with the same digital signature, allowing an attacker to masquerade as the affected service	**7.5**

Recommendations: Contact the Certificate Authority to have the SSL certificate

reissued

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium-strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
CVE-2001-1473, CVE-2001-0572, CVE-2001-0361	The remote SSH daemon supports connections made using the version 1.33 and/or 1.5 of the SSH protocol	**7.3**

Recommendations: Disable compatibility with version 1 of the SSH protocol

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
Unencrypted Telnet Server	The remote host is running a Telnet server over an unencrypted channel	**6.5**

Recommendations: Disable the Telnet service and use SSH instead

Vulnerability	Description	Score
CVE-2015-2808, CVE-2013-2566	The remote host supports the use of RC4 in one or more cipher suites. The RC4 cipher is flawed in its generation of a pseudo-random stream of bytes so that a wide variety of small biases are introduced into the stream, decreasing its randomness	**5.9**

Recommendations: Reconfigure the affected application, if possible, to avoid use of RC4 ciphers. Consider using TLS 1.2 with AES-GCM suites subject to browser and web server support

172.16.34.209- Palo Alto Networks

Critical	High	Medium	Low	Info
0	1	6	4	39

Vulnerability	Description	Score
CVE-2016-2183	The remote host supports the use of SSL ciphers that offer medium-strength encryption	**7.5**

Recommendations: Reconfigure the affected application if possible to avoid use of medium strength ciphers

Vulnerability	Description	Score
SSL Certificate Cannot Be Trusted	The web page SSL certificate cannot be trusted. This leaves the website vulnerable to a man-in-the-middle attack and data breaches and allows attackers to intercept sensitive user data	**6.5**

Recommendations: Purchase or generate a proper SSL certificate for the service

Vulnerability	Description	Score
CWE-327	The remote service accepts connections encrypted using TLS 1.1 TLS 1.1 lacks support for the current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1	**6.5**

Recommendations: Enable support for TLS 1.2 and/or 1.3 an disable support for TLS 1.1

Vulnerability	Description	Score
CVE-2020-11023, CVE-2020-11022	According to the self-reported version in the script, the version of JQuery hosted on the remote web server is greater than or equal to 1.2 and prior to 3.5.0. It is, therefore, affected by multiple cross site scripting vulnerabilities	**6.1**

Recommendations: Upgrade to JQuery version 3.5.0 or later

Vulnerability	Description	Score
CVE-2023-48795	The remote SSH server is vulnerable to a man-in-the-middle prefix truncation weakness known as Terrapin. This can allow a remote, man-in-the-middle attacker to bypass integrity checks and downgrade the connection's security	**5.9**

Recommendations: Contact the vendor for an update with the strict key exchange countermeasures or disable the affected algorithms

Vulnerability	Description	Score
SSH Weak Algorithms Supported	The remote SSH Server is configured to use the Arcfour stream cipher or no cipher at all. RFC 4253 advises against using Arcfour due to an issue with weak keys	**4.3**

Recommendations: Contact the vendor or consult product documentation to remove the weak ciphers

Appendix:

Nmap scan for both subnets:

https://drive.google.com/drive/folders/1ILX2OR96y_5M1VBmRBB7QHkW5RhWCwdD?usp=sharing

Greenbone Information:

https://drive.google.com/file/d/1iFgPnMpiOzWdAUkTe-643PwbVLWXOylG/view?usp=sharing

Nessus Information:

https://drive.google.com/file/d/1ggnbq-uW18BOSZkR-Os5--kx1PS-kBoe/view?usp=sharing

https://drive.google.com/file/d/1OFusp9hxz2df_ZrodHHiUkIZcqCYyOu3/view?usp=sharing

Shodan Information:

General Information

Country	**United States**
City	**Denver**
Organization	**Colorado Community College System**
ISP	**Colorado Community College and Occupational Education System**
ASN	**AS14737**

IPvoid Information:

IP Reputation API

164.47.47.11 Whois Check

```
NetRange:       164.47.0.0 - 164.47.255.255
CIDR:           164.47.0.0/16
NetName:        COEDS-NET
NetHandle:      NET-164-47-0-0-1
Parent:         NET164 (NET-164-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Colorado Community College System (CCCS-
RegDate:        1992-11-18
Updated:        2023-05-08
Ref:            https://rdap.arin.net/registry/ip/164.47


OrgName:        Colorado Community College System
OrgId:          CCCS-3-Z
Address:        1059 Alton Way
City:           Denver
StateProv:      CO
PostalCode:     80230
Country:        US
RegDate:        2023-04-25
Updated:        2023-04-25
Ref:            https://rdap.arin.net/registry/entity/CC


OrgAbuseHandle: HOSTM633-ARIN
OrgAbuseName:   Hostmaster
OrgAbusePhone:  +1-720-858-2821
OrgAbuseEmail:  Katherine.Stevenson@cccs.edu
OrgAbuseRef:    https://rdap.arin.net/registry/entity/HO
```

New Tools | Popular

- Make URL Harmless
- Sum of a List of Numbers
- Replace Commas with New Li
- Replace New Lines with Comm
- JSON Minify
- TLS Checker

NetApp
DRIVE THE FUTURE OF DATA AT NETAPP
Explore jobs →

Lead, learn, and grow as a nurse
Explore Careers
KAISER PERMANENTE careers

IP Reputation API

IPv4 CIDR Calculator

This online IPv4 CIDR calculator can convert an IPv4 CIDR address (i.e 73.35.0.0/20) to IP range. Calculate the first and last IP address in the CIDR range, the number of IPv4 addresses contained in the CIDR, and the netmask. Just enter the IPv4 CIDR address in the form below:

164.47.47.11 Get IP Range

```
Start IP: 164.47.47.11
End IP: 164.47.47.11
Number of IPs: 1
Subnet Range: 255.255.255.255
Wildcard: 0.0.0.0
```

Search Tools..

- Threat Intelligence APIs
- Harden Windows 10/11
- File Lines Manipulator
- Track USB Events

New Tools | Popular

- Make URL Harmless
- Sum of a List of Numbers
- Replace Commas with New Li
- Replace New Lines with Comm
- JSON Minify
- TLS Checker

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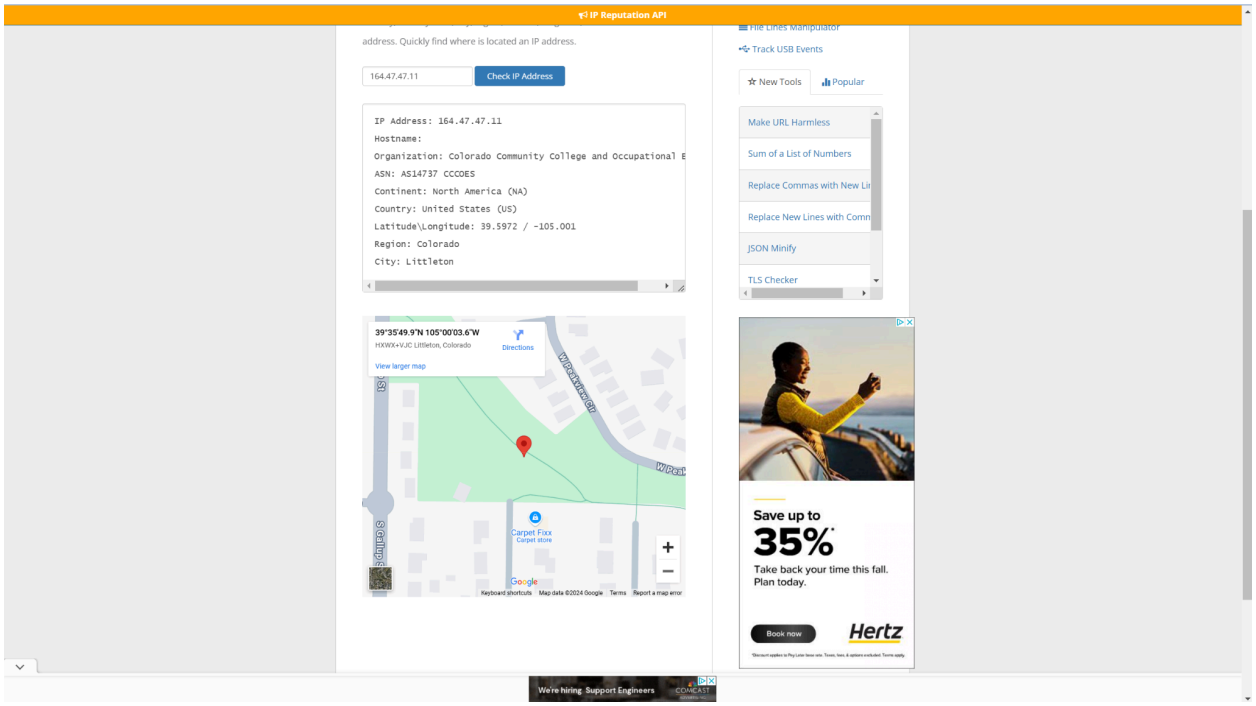
IP Reputation API
address. Quickly find where is located an IP address.
164.47.47.11
Check IP Address
IP Address: 164.47.47.11
Hostname:
Organization: Colorado Community College and Occupational E
ASN: AS14737 CCCOES
Continent: North America (NA)
Country: United States (US)
Latitude\Longitude: 39.5972 / -105.001
Region: Colorado
City: Littleton
39°35'49.9"N 105°00'03.6"W
HXWX+VJC Littleton, Colorado
Directions
View larger map
Carpet Fixx
Carpet store
Google
File Lines Manipulator
Track USB Events
New Tools
Popular
Make URL Harmless
Sum of a List of Numbers
JSON Minify
TLS Checker
Save up to
35%
Take back your time this fall.
Plan today.
Book now
Hertz
We're hiring Support Engineers
COMCAST